As usual, you'll find the code in the book's code repository on GitHub: https://github.com/AWSinAction/code3. Switch to the chapter06 directory, which includes all files needed for this example.

Time to write some Python code! Listing 6.4 for `lambda_function.py` shows the function, which receives an event from CloudTrail indicating that an EC2 instance has been launched recently, and adds a tag including the name of the instance's owner. The AWS SDK for Python 3.9, named `boto3`, is provided out of the box in the Lambda runtime environment for Python 3.9. In this example, you are using the AWS SDK to create a tag for the EC2 instance `ec2.create_tags(...)`. See the Boto3 documentation at https://boto3.readthedocs.io/en/latest/index.xhtml if you are interested in the details of `boto3`.

Listing 6.4 Lambda function adding a tag to EC2 instance

```
import boto3
ec2 = boto3.client('ec2')                                          ①

def lambda_handler(event, context):                                ②
  print(event)                                                     ③
  if "/" in event['detail']['userIdentity']['arn']:                ④
      userName = event['detail']['userIdentity']['arn'].split('/')[1]
  else:
      userName = event['detail']['userIdentity']['arn']
  instanceId = event['detail']['responseElements']['instancesSet']
  ➥ ['items'][0]['instanceId']                                    ⑤
  print("Adding owner tag " + userName + " to instance " + instanceId + ".")
  ec2.create_tags(Resources=[instanceId,],
  ➥ Tags=[{'Key': 'Owner', 'Value': userName},])                  ⑥
  return
```

① Creates an AWS SDK client for EC2

② The name of the function used as entry point for the Lambda function

③ Logs the incoming event for debugging

④ Extracts the user's name from the CloudTrail event